

南方硅谷 SV6358 系列芯片 NimBle 使用说明

简介

文档主要介绍 6358 系列芯片 nimble 功能的使用。

发布说明：

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1. 驱动编译和加载

- 修改 config.mak 编译选项，如下：



```
# TX_MULTI_URB 開啟或關閉
# 此功能不能與HCI TX AGGR共存
# 需要設定ssv6x5x-wifi.cfg 中 hw_cap_hci_tx_aggr = off
CONFIG_SSV_MULTI_URB ?= n
ifeq ($(CONFIG_SSV_MULTI_URB), y)
ccflags-y += -DCONFIG_USB_TX_MULTI_URB
endif

## FMAC使用
ccflags-y += -DCONFIG_FMAC
# OBSS scans and generate 20/40 BSS coex report
#ccflags-y += -DNEED_OBSS_SCAN
# send keep alive frame in driver side every 30 seconds
#ccflags-y += -DSEND_KEEP_ALIVE

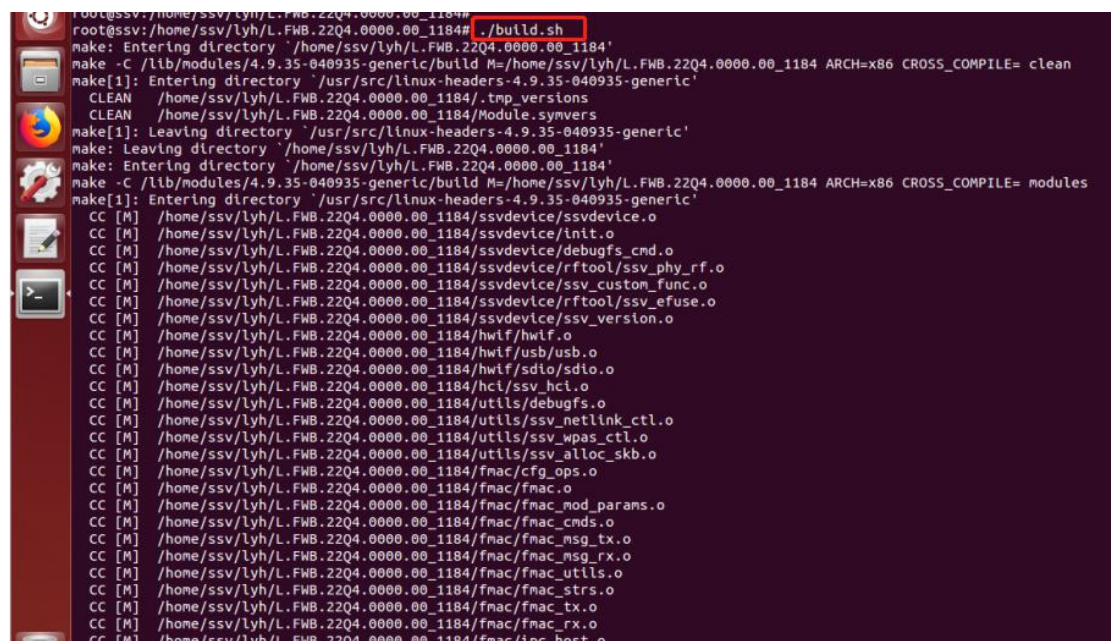
## NIMBLE使用((注意与BLE的使用为2选1))
ccflags-y += -DCONFIG_NIMBLE

## BLE HCI usage (注意与NIMBLE的使用为2选1)
#CONFIG_BLE ?= n
#ccflags-$(CONFIG_BLE) += -DCONFIG_BLE

# BLE HCI BUS 取值 :
# CONFIG_BLE_HCI_OVER_UART:0
# CONFIG_BLE_HCI_OVER_HWIF:1 USB or SDIO, base on HWIF
ccflags-y += -DCONFIG_BLE_HCI_BUS=1

## wpa_supplicant接口使用
CONFIG_WPA_SUPPLICANT_CTL ?= y
ccflags-$(CONFIG_WPA_SUPPLICANT_CTL) += -DCONFIG_WPA_SUPPLICANT_CTL
```

- 编译驱动如下：



```
root@ssv:/home/ssv/lyh/L.FWB.22Q4.0000.00_1184# ./build.sh
make: Entering directory `/home/ssv/lyh/L.FWB.22Q4.0000.00_1184'
make -C /lib/modules/4.9.35-040935-generic/build M=/home/ssv/lyh/L.FWB.22Q4.0000.00_1184 ARCH=x86 CROSS_COMPILE= clean
make[1]: Entering directory `/usr/src/linux-headers-4.9.35-040935-generic'
CLEAN /home/ssv/lyh/L.FWB.22Q4.0000.00_1184/.tmp_versions
CLEAN /home/ssv/lyh/L.FWB.22Q4.0000.00_1184/Module.symvers
make[1]: Leaving directory `/usr/src/linux-headers-4.9.35-040935-generic'
make: Leaving directory `/home/ssv/lyh/L.FWB.22Q4.0000.00_1184'
make: Entering directory `/home/ssv/lyh/L.FWB.22Q4.0000.00_1184'
make -C /lib/modules/4.9.35-040935-generic/build M=/home/ssv/lyh/L.FWB.22Q4.0000.00_1184 ARCH=x86 CROSS_COMPILE= modules
make[1]: Entering directory `/usr/src/linux-headers-4.9.35-040935-generic'
CC [M] /home/ssv/lyh/L.FWB.22Q4.0000.00_1184/ssvdevice/ssvdevice.o
CC [M] /home/ssv/lyh/L.FWB.22Q4.0000.00_1184/ssvdevice/init.o
CC [M] /home/ssv/lyh/L.FWB.22Q4.0000.00_1184/ssvdevice/debugfs_cmd.o
CC [M] /home/ssv/lyh/L.FWB.22Q4.0000.00_1184/ssvdevice/rftool/ssv_phy_rf.o
CC [M] /home/ssv/lyh/L.FWB.22Q4.0000.00_1184/ssvdevice/ssv_custom_func.o
CC [M] /home/ssv/lyh/L.FWB.22Q4.0000.00_1184/ssvdevice/rftool/ssv_efuse.o
CC [M] /home/ssv/lyh/L.FWB.22Q4.0000.00_1184/ssvdevice/ssv_version.o
CC [M] /home/ssv/lyh/L.FWB.22Q4.0000.00_1184/hwlf/hwlf.o
CC [M] /home/ssv/lyh/L.FWB.22Q4.0000.00_1184/hwlf/usb/usb.o
CC [M] /home/ssv/lyh/L.FWB.22Q4.0000.00_1184/hwlf/sdio/sdio.o
CC [M] /home/ssv/lyh/L.FWB.22Q4.0000.00_1184/hci/ssv_hci.o
CC [M] /home/ssv/lyh/L.FWB.22Q4.0000.00_1184/utlis/debugfs.o
CC [M] /home/ssv/lyh/L.FWB.22Q4.0000.00_1184/utlis/ssv_netlink_ctl.o
CC [M] /home/ssv/lyh/L.FWB.22Q4.0000.00_1184/utlis/ssv_wpas_ctl.o
CC [M] /home/ssv/lyh/L.FWB.22Q4.0000.00_1184/utlis/ssv_alloc_skb.o
CC [M] /home/ssv/lyh/L.FWB.22Q4.0000.00_1184/fmac/cfg_ops.o
CC [M] /home/ssv/lyh/L.FWB.22Q4.0000.00_1184/fmac/fmac.o
CC [M] /home/ssv/lyh/L.FWB.22Q4.0000.00_1184/fmac/fmac_mod_params.o
CC [M] /home/ssv/lyh/L.FWB.22Q4.0000.00_1184/fmac/fmac_cmds.o
CC [M] /home/ssv/lyh/L.FWB.22Q4.0000.00_1184/fmac/fmac_msg_tx.o
CC [M] /home/ssv/lyh/L.FWB.22Q4.0000.00_1184/fmac/fmac_msg_rx.o
CC [M] /home/ssv/lyh/L.FWB.22Q4.0000.00_1184/fmac/fmac_utlis.o
CC [M] /home/ssv/lyh/L.FWB.22Q4.0000.00_1184/fmac/fmac_strs.o
CC [M] /home/ssv/lyh/L.FWB.22Q4.0000.00_1184/fmac/fmac_tx.o
CC [M] /home/ssv/lyh/L.FWB.22Q4.0000.00_1184/fmac/fmac_rx.o
CC [M] /home/ssv/lyh/L.FWB.22Q4.0000.00_1184/fmac/ipc_host.o
```

- 修改 ssv6x5x-wifi.cfg，配置 nimble fw，如下：

```
../L.FWB.22Q4.0000.00_1184# vim ssv6x5x-wifi.cfg
#####
#firmware_path = /home/xxx/xxx/image/
firmware_path = /mnt/hgfs/E/EFWB/L.FWB.22Q4.0000.00_1184/image/ 设置nimble fw路径
#####
# Firmware name
#
# 若有設定firmware path的話才有用，內部預設firmware name使用"ssv6x5x-sw.bin"
#
#####
#firmware name = ssv6x5x-sw.bin
firmware_name = ssv6x5x-sw_nimble.bin 设置nimble fw名称
#####
# Firmware choice (for firmware setting with priority 3. default firmware)
#
# 1: Normal mode (ssv6x5x-sw.h) [Default]
# others: Engineer mode (ssv6x5x-sw_2.h)
#
#####
#firmware_choice = 2 默认关闭或选择1
#####
# Flow control setting (TX)
```

图 1. 配置 nimble fw 说明

- 加载驱动并使对应 wifi 接口 up 起来:

```
root@ubuntu:/mnt/hgfs/E/EFWB/L.FWB.22Q4.0000.00_1184# ls
ap_launch.sh  build_t31.sh  build.sh  config.mak  hostapd.conf  Kconfig  load.sh  Makefile  modules.order  Module.symvers  ssv6x5x.mod.c  ssv6x5x.mod.o  ssv6x5x.o  ssv6x5x-wifi.cfg
ap_shutdown.sh  cli  build.sh  config.mak  hostapd.conf  Kconfig  load.sh  Makefile  modules.order  Module.symvers  ssv6x5x-generic-wlan.c  ssv6x5x-generic-wlan.o  ssv6x5x-wifi.cfg
root@ubuntu:/mnt/hgfs/E/EFWB/L.FWB.22Q4.0000.00_1184# insmod ssv6x5x.ko stacfgpath=ssv6x5x-wifi.cfg
root@ubuntu:/mnt/hgfs/E/EFWB/L.FWB.22Q4.0000.00_1184# ifconfig
ens33  Link encap:Ethernet HWaddr 00:0c:29:dc:25:8e
        inet addr:10.10.20.91 Bcast:10.10.20.255 Mask:255.255.255.0
        inet6 addr: fe80::7762:e134:8be7:8f6/64 Scope:Link
        UP BROADCAST RUNNING MULTICAST  MTU:1500  Metric:1
        RX packets:3042265 errors:0 dropped:2 overruns:0 frame:0
        TX packets:9325 errors:0 dropped:0 overruns:0 carrier:0
        collisions:0 txqueuelen:1000
        RX bytes:376684282 (376.6 MB)  TX bytes:776728 (776.7 KB)

lo  Link encap:Local Loopback
        inet addr:127.0.0.1 Mask:255.0.0.0
        inet6 addr: ::1/128 Scope:Host
        UP LOOPBACK RUNNING  MTU:65536  Metric:1
        RX packets:1220 errors:0 dropped:0 overruns:0 frame:0
        TX packets:1220 errors:0 dropped:0 overruns:0 carrier:0
        collisions:0 txqueuelen:1000
        RX bytes:97221 (97.2 KB)  TX bytes:97221 (97.2 KB)

wlx002345678920 Link encap:Ethernet HWaddr 00:23:45:67:89:20
        UP BROADCAST MULTICAST  MTU:1500  Metric:1
        RX packets:0 errors:0 dropped:0 overruns:0 frame:0
        TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
        collisions:0 txqueuelen:1000
        RX bytes:0 (0.0 B)  TX bytes:0 (0.0 B)
```


2.编译 ble_app 应用程序

- 如下，编译应用 ble_app 应用程序：

```
root@ssv:/home/ssv/lyh/L.FWB.22Q4.0000.00_1184/tools/ble/ble_app#
root@ssv:/home/ssv/lyh/L.FWB.22Q4.0000.00_1184/tools/ble/ble_app# make clean;make
rm -f *.o ./cli/*.o ./services/api/ssv_netlink/*.o ./services/api/ -o ble_app
gcc -c -Wall -fshort-enum -I./services/api/ssv_netlink -I./services/api -I./cli -I./services/include/apache-mynew-nimble-1.2.0/host/bluedroid
/apl/include -I./services/include/apache-mynew-nimble-1.2.0/common/include main.c -o main.o
gcc -c -Wall -fshort-enum -I./services/api/ssv_netlink -I./services/api -I./cli -I./services/include/apache-mynew-nimble-1.2.0/host/bluedroid
/apl/include -I./services/include/apache-mynew-nimble-1.2.0/common/include services/api/ssv_netlink/ssv_ble_services.c -o services/api/ssv_ne
tlink/ssv_ble_services.o
gcc -c -Wall -fshort-enum -I./services/api/ssv_netlink -I./services/api -I./cli -I./services/include/apache-mynew-nimble-1.2.0/host/bluedroid
/apl/include -I./services/include/apache-mynew-nimble-1.2.0/common/include services/api/ssv_netlink/ssv_netlink.c -o services/api/ssv_netlink
/ssv_netlink.o
gcc -c -Wall -fshort-enum -I./services/api/ssv_netlink -I./services/api -I./cli -I./services/include/apache-mynew-nimble-1.2.0/host/bluedroid
/apl/include -I./services/include/apache-mynew-nimble-1.2.0/common/include services/api/ssv_hal_ble_common.c -o services/api/ssv_hal_ble_comm
on.o
gcc -c -Wall -fshort-enum -I./services/api/ssv_netlink -I./services/api -I./cli -I./services/include/apache-mynew-nimble-1.2.0/host/bluedroid
/apl/include -I./services/include/apache-mynew-nimble-1.2.0/common/include services/api/ssv_hal_ble_gap.c -o services/api/ssv_hal_ble_gap.o
gcc -c -Wall -fshort-enum -I./services/api/ssv_netlink -I./services/api -I./cli -I./services/include/apache-mynew-nimble-1.2.0/host/bluedroid
/apl/include -I./services/include/apache-mynew-nimble-1.2.0/common/include services/api/ssv_hal_ble_gatts.c -o services/api/ssv_hal_ble_gatts
.o
gcc -c -Wall -fshort-enum -I./services/api/ssv_netlink -I./services/api -I./cli -I./services/include/apache-mynew-nimble-1.2.0/host/bluedroid
/apl/include -I./services/include/apache-mynew-nimble-1.2.0/common/include services/api/ssv_hal_ble_gattc.c -o services/api/ssv_hal_ble_gattc
.o
gcc -c -Wall -fshort-enum -I./services/api/ssv_netlink -I./services/api -I./cli -I./services/include/apache-mynew-nimble-1.2.0/host/bluedroid
/apl/include -I./services/include/apache-mynew-nimble-1.2.0/common/include services/api/ssv_gatt_common.c -o services/api/ssv_gatt_common
.o
gcc -c -Wall -fshort-enum -I./services/api/ssv_netlink -I./services/api -I./cli -I./services/include/apache-mynew-nimble-1.2.0/host/bluedroid
/apl/include -I./services/include/apache-mynew-nimble-1.2.0/common/include services/api/ssv_gap_ble_api.c -o services/api/ssv_gap_ble_api.o
gcc -c -Wall -fshort-enum -I./services/api/ssv_netlink -I./services/api -I./cli -I./services/include/apache-mynew-nimble-1.2.0/host/bluedroid
/apl/include -I./services/include/apache-mynew-nimble-1.2.0/common/include cli/cli_key.c -o cli/cli_key.o
gcc -c -Wall -fshort-enum -I./services/api/ssv_netlink -I./services/api -I./cli -I./services/include/apache-mynew-nimble-1.2.0/host/bluedroid
/apl/include -I./services/include/apache-mynew-nimble-1.2.0/common/include cli/cli.c -o cli/cli.o
make: Circular ble_app <- ble_app dependency dropped.
gcc -I./services/api/ssv_netlink -I./services/api -I./cli -I./services/include/apache-mynew-nimble-1.2.0/host/bluedroid/apl/include -I./servic
es/include/apache-mynew-nimble-1.2.0/common/include main.o ./services/api/ssv_netlink/ssv_ble_services.o ./services/api/ssv_netlink/ssv_netli
nk.o ./services/api/ssv_hal_ble_common.o ./services/api/ssv_hal_ble_gap.o ./services/api/ssv_hal_ble_gatts.o ./services/api/ssv_hal_ble_gattc.o
./services/api/ssv_gatt_common.o ./services/api/ssv_gap_ble_api.o ./cli/cli_key.o ./cli/cli.o -lpthread -o ble_app -ldl
root@ssv:/home/ssv/lyh/L.FWB.22Q4.0000.00_1184/tools/ble/ble_app# ls
ble_app  cli  main.c  main.o  Makefile  services
root@ssv:/home/ssv/lyh/L.FWB.22Q4.0000.00_1184/tools/ble/ble_app#
```

3. 使用 nimble 功能

3.1. 运行 ble_app 广播蓝牙名称:

```

root@ssv:/home/ssv/lyh/L.FWB.22Q4.0000.00_1184/tools/ble/ble_app# ls
ble_app cli main.c Makefile services
root@ssv:/home/ssv/lyh/L.FWB.22Q4.0000.00_1184/tools/ble/ble_app#
root@ssv:/home/ssv/lyh/L.FWB.22Q4.0000.00_1184/tools/ble/ble_app# ./ble_app

ble> init=1
bleinit=1

ble> nla_type : 6
evtid:0x10000 datalen: 4
SSV_GATTS_REG_EVT

ble> addservice

ble> nla_type : 6
evtid:0x10016 datalen: 40
SSV_GATTS_CREAT_ATTR_TAB_EVT num_handle:6
a b c d e f

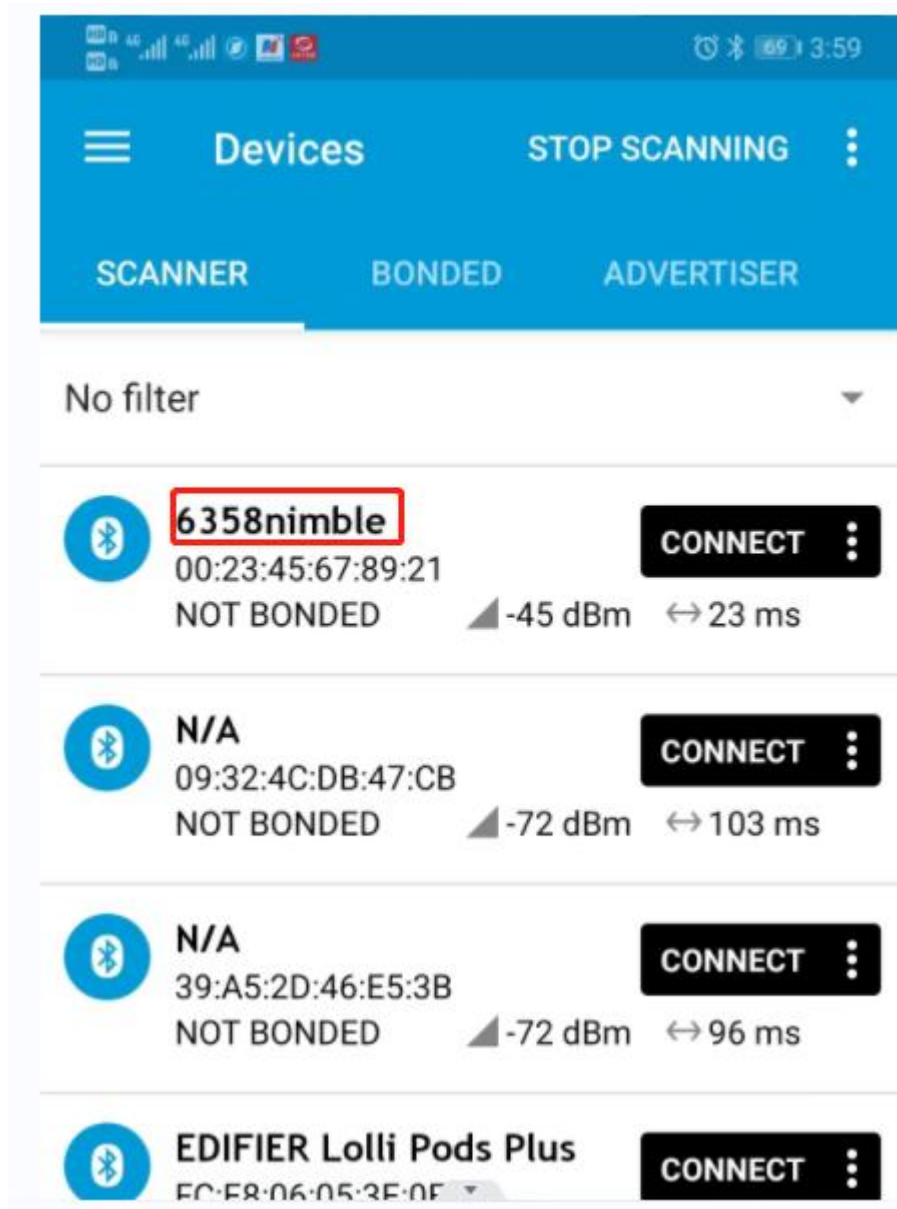
ble>
ble> advert=1,6358nimble
bleADV=1,6358nimble

ble> nla_type : 6
evtid:0x4 datalen: 1
SSV_GAP_BLE_ADV_DATA_RAW_SET_COMPLETE_EVT
nla_type : 6
evtid:0x5 datalen: 1
SSV_GAP_BLE_SCAN_RSP_DATA_RAW_SET_COMPLETE_EVT
nla_type : 6
evtid:0x6 datalen: 1
SSV_GAP_BLE_ADV_START_COMPLETE_EVT

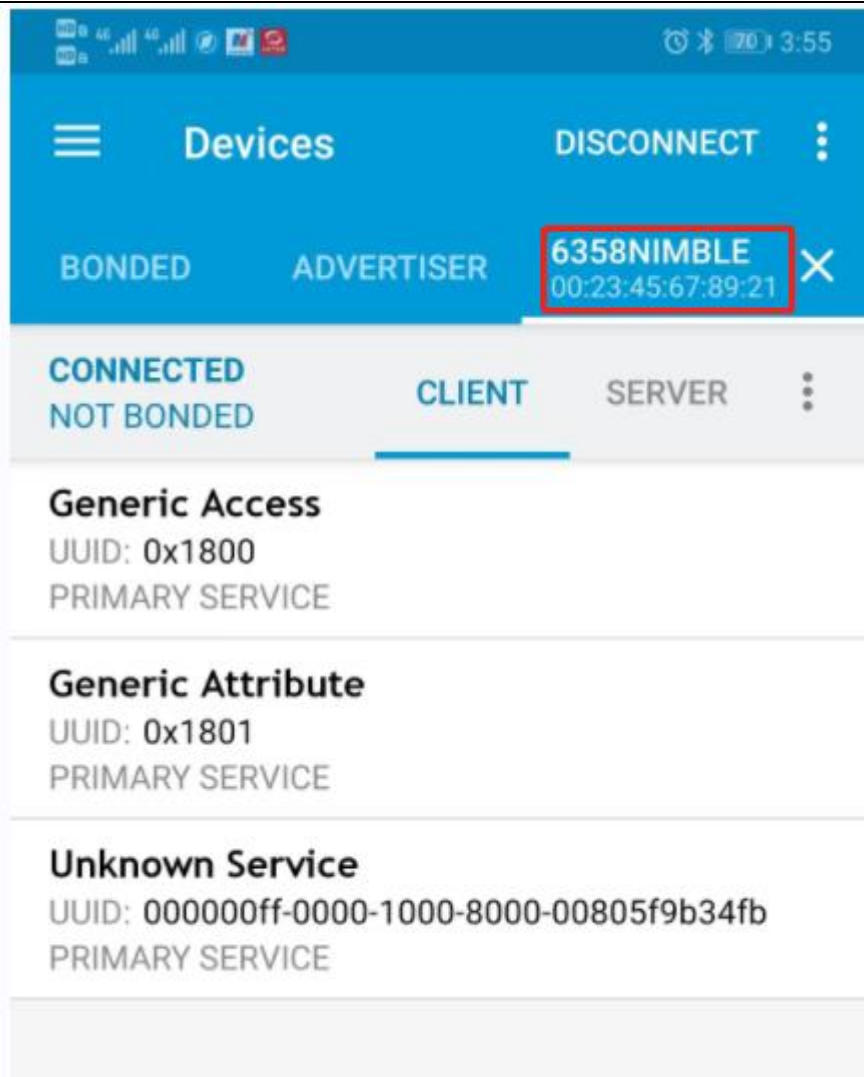
```

3.2. 使用手机 app 扫描到 Ble 名称并 connect

- 扫描并进行连接:



- Connect 后显示如下:

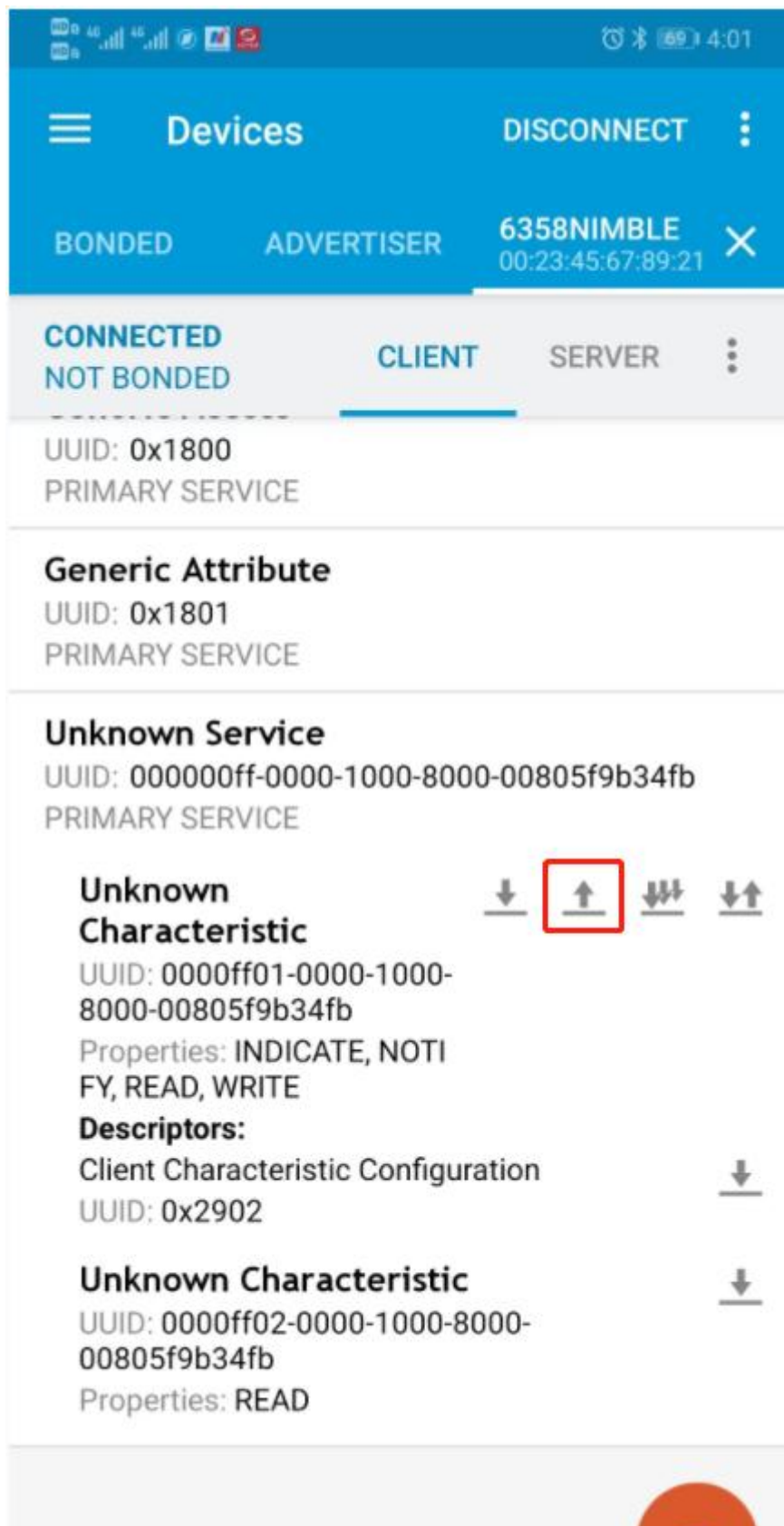


- 连接成功 6358 端信息:

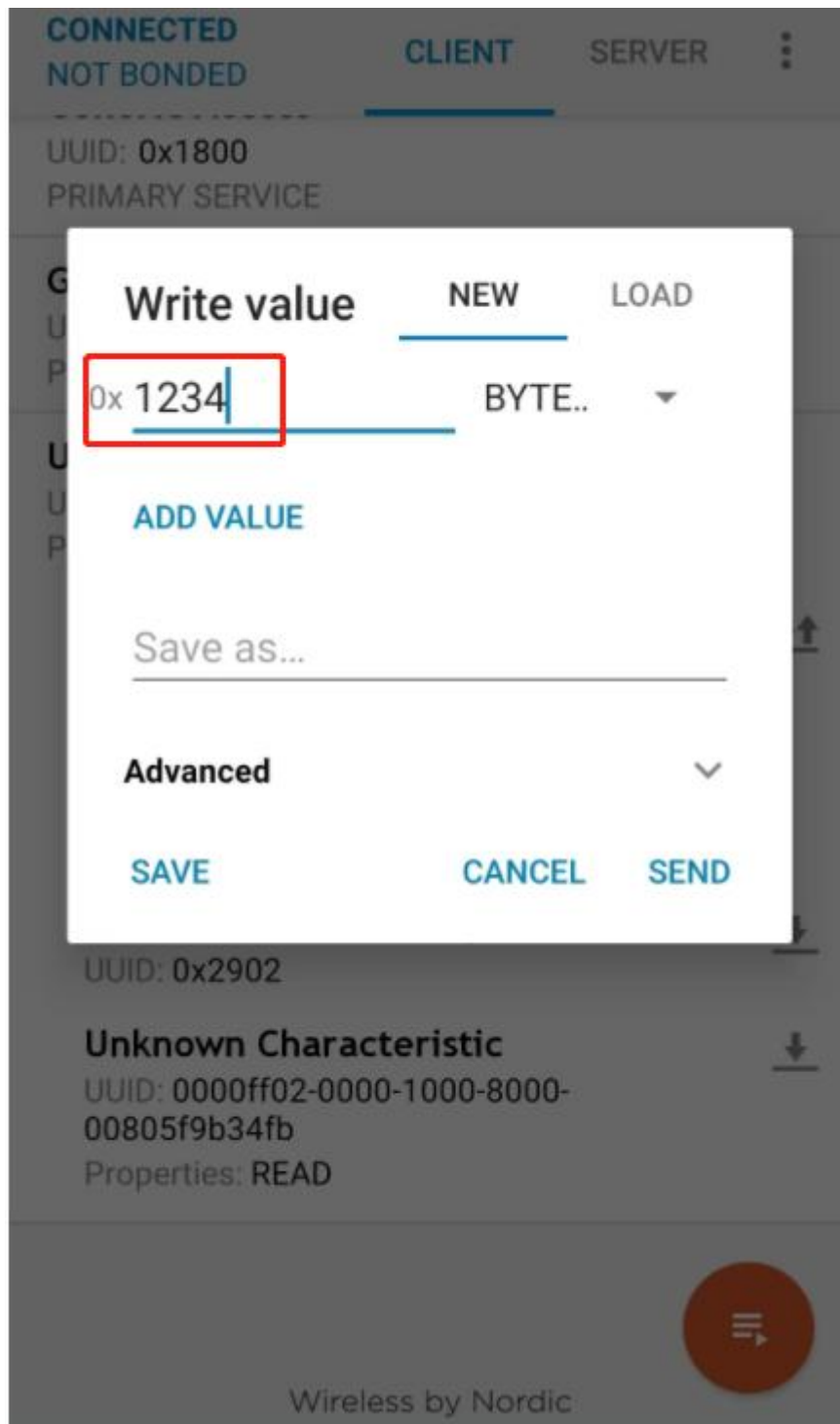
```
ble>
ble> nla_type : 6
evtid:0x1000e datalen: 14
Connect id=3
Remote BD=7E:9E:7C:5E:ED:9E
nla_type : 6
evtid:0x10004 datalen: 4
SSV_GATTS_MTU_EVT
nla_type : 6
evtid:0x14 datalen: 18
SSV_GAP_BLE_UPDATE_CONN_PARAMS_EVT
nla_type : 6
evtid:0x14 datalen: 18
SSV_GAP_BLE_UPDATE_CONN_PARAMS_EVT
ble>
```

3.3. Nimble 接收数据

- 手机 app 端触发发送:



- 填写发送数据:

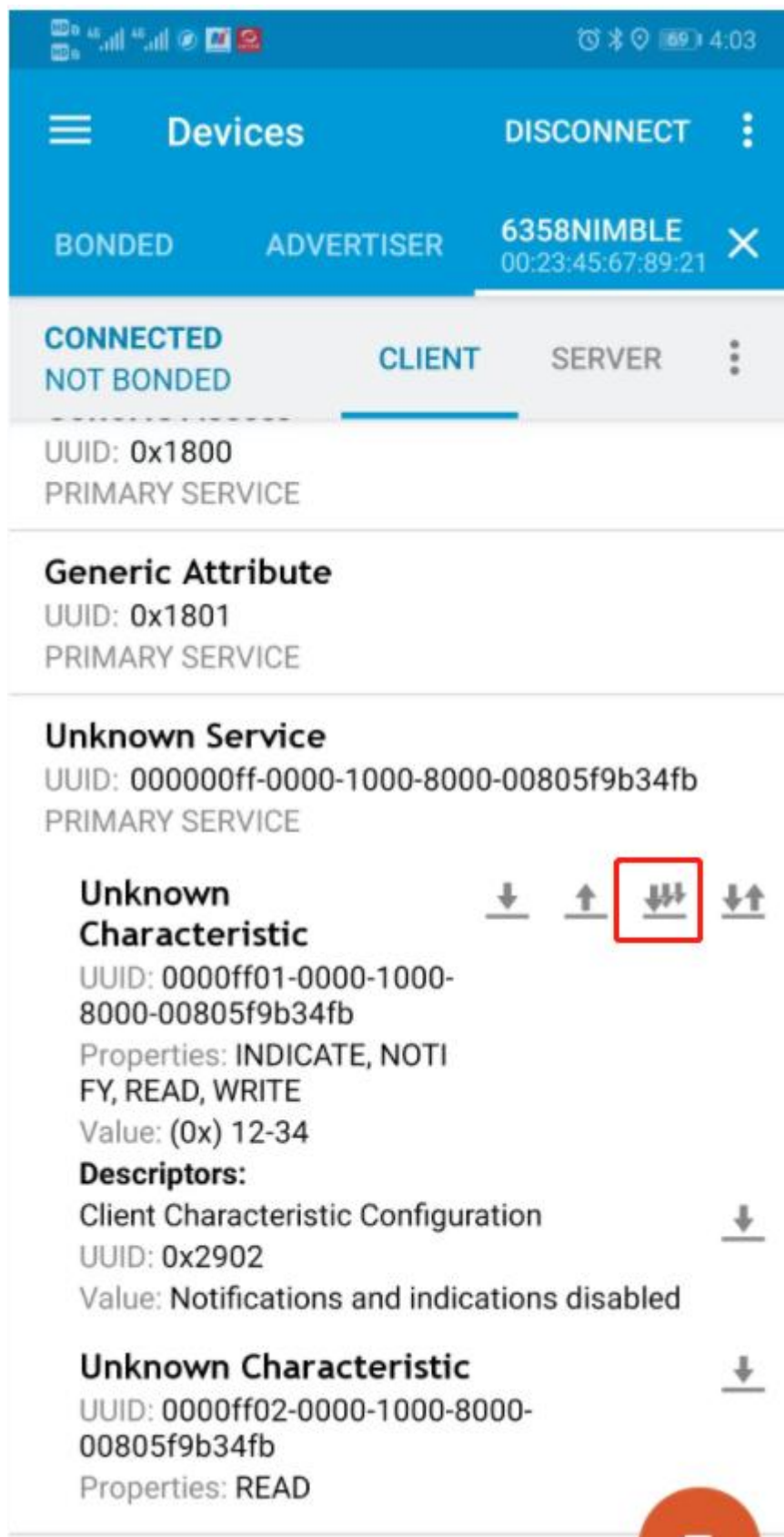


- 6358 端收到数据：

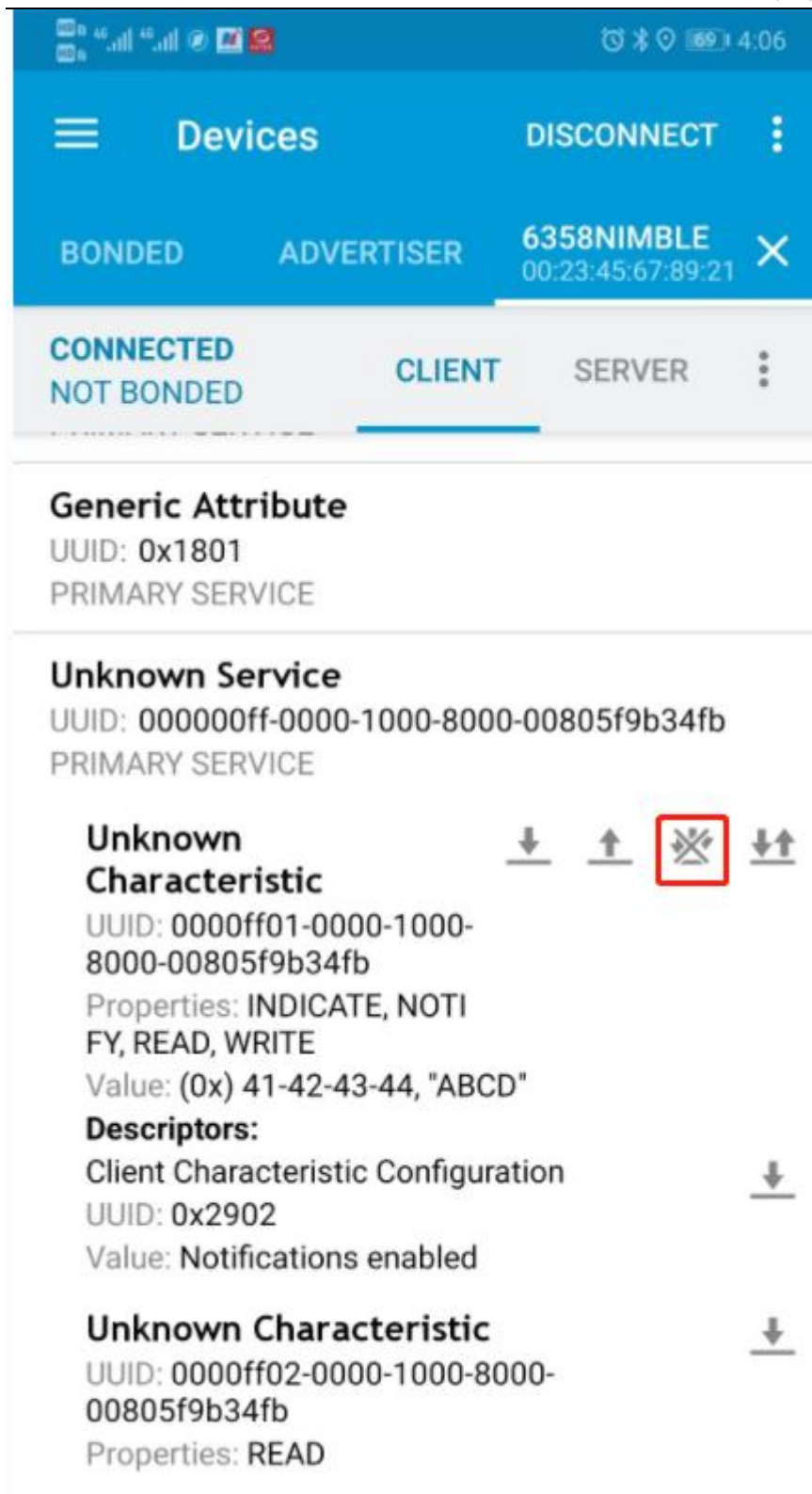
```
ble>
ble> nla_type : 6
evtid:0x10002 datalen: 30
SSV_GATTS_WRITE_EVT handle:12, len:2, rsp:0
recv: 12 34 0 0
ble>
```

3.4. 发送数据

- 手机 app 触发接收:



- 显示下图表明已处于接收状态：



- 6358 端查看前面广播数据，确定属性值：

```
ble> addservice

ble> nla_type : 6
evtid:0x10016 datalen: 40
SSV_GATTS_CREAT_ATTR_TAB_EVT num_handle:6
a b c d e f
ble>
```

 12 (十六进制变成十进制)

- 6358 端根据属性值发送数据 “ABCD” :

```
ble>
ble> send=12,ABCD

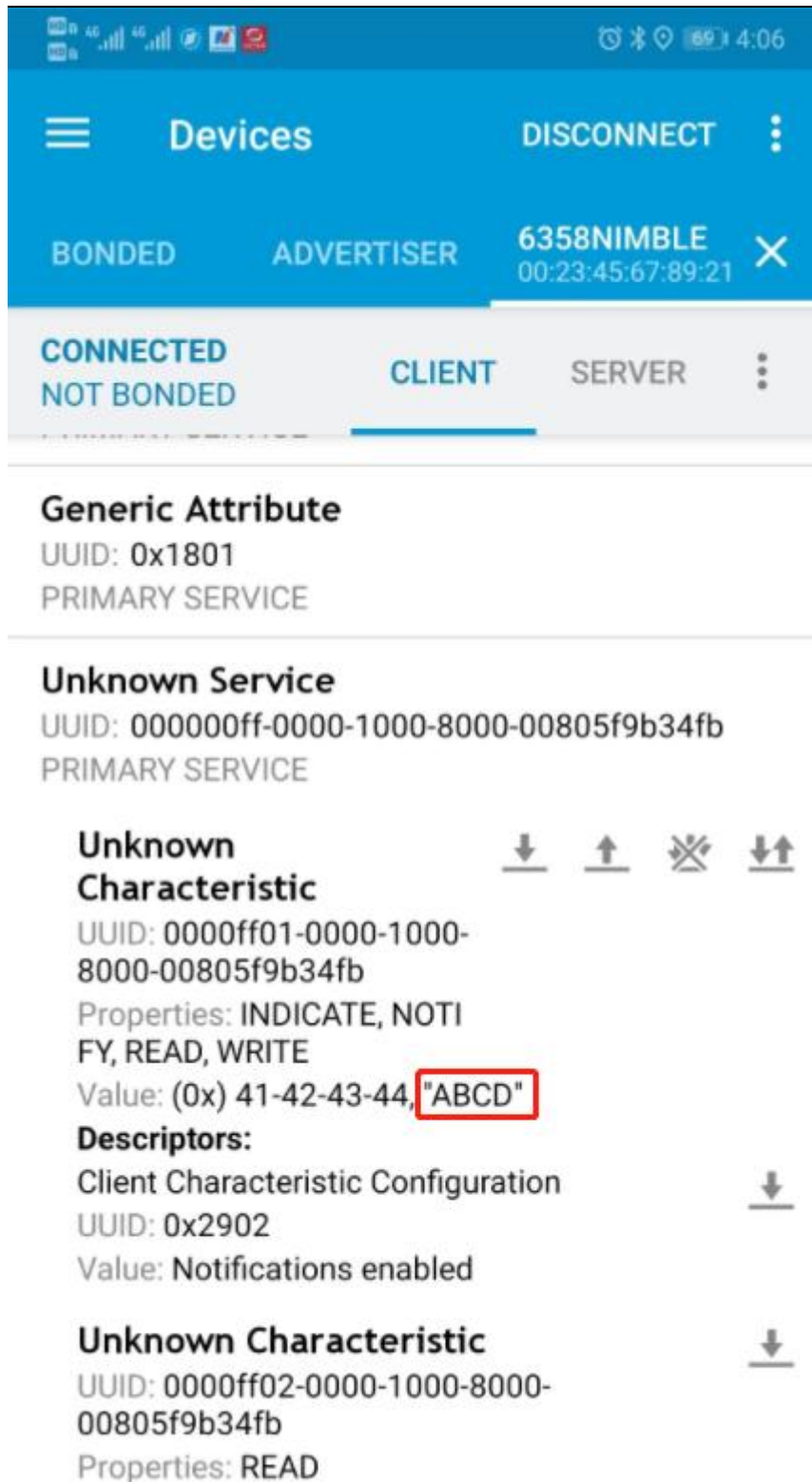
ble> nla_type : 6
evtid:0x10005 datalen: 28
SSV_GATTS_CONF_EVT

ble>
ble>
ble> send=12,ABCD

ble> nla_type : 6
evtid:0x10005 datalen: 28
SSV_GATTS_CONF_EVT

ble>
```

- 手机端显示收到数据:



4. 正常功能（normal fw）模式

要保持 wifi 高吞吐等性能，nimble 功能使用结束后，驱动需要切换正常功能（normal fw）模式。ssv6x5x-wifi.cfg 改回默认值，重新加载驱动。

```
#####
# Firmware setting
#
# Priority.1 insmod parameter "cfgfirmwarepath" (insmod的時候帶入)
# Priority.2 firmware_path (在下面設定)
# Priority.3 default firmware (前兩個都沒設定會load firmware header)
#
#####
#firmware_path = /home/xxx/xxx/image/
#firmware_path = /home/ssv/lyh/L.FWB.22Q4.0000.00_1184/image/
#
#####
# Firmware name
#
# 若有設定firmware path的話才有用，內部預設firmware name使用"ssv6x5x-sw.bin"
#
#####
#firmware_name = ssv6x5x-sw.bin
#firmware_name = ssv6x5x-sw_nimble.bin
#
#####
# Firmware choice (for firmware setting with priority 3. default firmware)
#
# 1: Normal mode (ssv6x5x-sw.h) [Default]
# others: Engineer mode (ssv6x5x-sw_2.h)
#
#####
#firmware_choice = 2
#
#####
# Flow control setting (TX)
```

以上，谢谢！